

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks: 40

Annexure A

Mock Interview

Code of Conduct:

I M. Madhulika (192472188) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Madhulika

QUESTIONS: 40

Total Marks: 40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)

7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)
27. How would you present your simulation results to a client? (BL4)(1)
28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect

Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. How does network congestion affect packet delay and packet loss?

Answer:

Network congestion occurs when too much traffic enters the network. It increases packet delay because packets wait in queues, and severe congestion can cause packet loss when buffers become full.

2. Star vs Mesh topology

Answer:

A star topology is easier to manage but depends on the central switch. A mesh topology provides higher reliability because multiple paths are available, but it is more expensive and complex.

3. Why do more routers increase end-to-end delay?

Answer:

Each router adds processing, queuing, transmission, and propagation delay. Therefore, more routers generally increase the total end-to-end delay.

4. How does QoS improve voice and video traffic?

Answer:

QoS gives priority to real-time traffic such as voice and video. It helps reduce delay, jitter, and packet loss, providing smoother communication.

5. Fiber optic vs copper cables

Answer:

Fiber optic provides higher speed, longer-distance communication, lower signal loss, and immunity to electromagnetic interference. Copper is generally cheaper but has lower performance over long distances.

6. How do VLANs improve performance and security?

Answer:

VLANs divide a network into smaller logical networks. This reduces unnecessary broadcast traffic and improves performance, management, and security.

7. TCP vs UDP

Answer:

TCP is reliable and connection-oriented, so it is suitable for web browsing and file transfer.

UDP is faster and connectionless, so it is useful for gaming, streaming, and VoIP.

8. Why is load balancing important?

Answer:

Load balancing distributes traffic across multiple servers or network paths. It prevents overload, improves performance, and increases availability and reliability.

9. Effect of frequent routing updates

Answer:

Frequent routing updates consume bandwidth and processing resources. Too many updates can reduce network efficiency, although timely updates help maintain accurate routing information.

10. Security vs network performance

Answer:

Firewalls and encryption improve security but require additional processing. Therefore, strong security can slightly reduce performance, so we need a balance between security and efficiency.

11. A network has high delay. How will you identify and fix it?

Answer:

I would check ping, traceroute, bandwidth utilization, congestion, and router load. Then I would remove bottlenecks, optimize routing, increase bandwidth, or reduce unnecessary traffic.

12. Throughput is low despite high bandwidth. Why?

Answer:

Possible reasons are congestion, packet loss, high latency, protocol overhead, hardware limitations, or poor network configuration.

13. How would you reduce packet loss?

Answer:

I would identify congestion or faulty links, increase bandwidth, optimize routing, improve network hardware, and apply QoS and traffic management.

14. VoIP call is breaking frequently. How will you troubleshoot?

Answer:

I would check packet loss, delay, jitter, bandwidth, and connectivity. Then I would prioritize VoIP traffic using QoS and fix congestion or faulty connections.

15. Network has high jitter. Improvements?

Answer:

I would reduce congestion, prioritize real-time traffic using QoS, improve routing, and provide sufficient bandwidth.

16. How do you simulate a congested network in Packet Tracer?

Answer:

I would create multiple devices and generate high traffic through a limited-bandwidth link. Then I would observe packet delay, packet loss, and throughput.

17. Packets are delayed in a router. What steps will you take?

Answer:

I would check the router's CPU utilization, queue size, interface utilization, routing table, and congestion. Then I would optimize the configuration or increase capacity.

18. How would you optimize a network for online gaming?

Answer:

I would reduce latency and jitter, minimize packet loss, use QoS to prioritize gaming traffic, and select efficient routing paths.

19. Design a small network and measure performance

Answer:

I would connect PCs through a switch and router in Packet Tracer. Then I would use ping and simulation tools to measure delay, packet loss, throughput, and RTT.

20. RTT is very high. How will you diagnose it?

Answer:

I would use ping and traceroute to identify where the delay occurs. Then I would check congestion, routing paths, link quality, and router performance.

21. Explain network performance to a non-technical person.

Answer:

Network performance means how fast and reliably data travels from one device to another. A good network delivers data quickly with minimum delay and loss.

22. Explain throughput with a real-life example.

Answer:

Throughput is the actual amount of data successfully transferred per second. For example, if a network successfully transfers 50 MB every second, its throughput is 50 MB/s.

23. How does packet loss affect video streaming?

Answer:

Packet loss means some video data does not reach the destination. This can cause **buffering**, freezing, reduced quality, or interruptions.

24. Explain delay types in under one minute.

Answer:

There are four main delays: processing delay, queuing delay, transmission delay, and propagation delay. Processing is time taken to examine a packet, queuing is waiting time, transmission is time to put the packet onto the link, and propagation is the time for the signal to travel through the medium.

25. Describe Cisco Packet Tracer professionally.

Answer:

Cisco Packet Tracer is a network simulation tool used to design, configure, and test computer networks virtually without requiring physical networking equipment.

26. Explain QoS in simple and technical terms.

Answer:

Simply, QoS means giving priority to important network traffic. Technically, QoS manages traffic to control delay, jitter, bandwidth, and packet loss, especially for real-time applications.

27. How would you present simulation results to a client?

Answer:

I would present the network topology, performance metrics, observations, problems, and proposed improvements using simple tables or graphs.

28. Explain jitter using a real-world analogy.

Answer:

Jitter is the variation in packet arrival time. For example, if a person receives voice messages at irregular intervals instead of evenly, the communication becomes disturbed.

29. How did you test network performance in a lab?

Answer:

I created the network in Cisco Packet Tracer, configured the devices, generated traffic, and measured parameters such as delay, packet loss, throughput, and RTT.

30. Importance of performance metrics

Answer:

Performance metrics help us understand the condition of a network. Bandwidth, throughput, delay, jitter, packet loss, and RTT help identify problems and improve network performance.

31. Introduce yourself and explain your knowledge of network performance.

Answer:

Good morning. I am a Computer Science Engineering student. I have basic knowledge of computer networks and network performance metrics such as bandwidth, throughput, delay, jitter, packet loss, and RTT. I have also practiced network simulation using Cisco Packet Tracer.

32. Why are you interested in networking and simulation tools?

Answer:

I am interested in networking because it helps me understand how devices communicate. Simulation tools like Packet Tracer allow me to practice network configuration and troubleshooting without physical hardware.

33. What makes you confident in analyzing network performance?

Answer:

"I understand the major performance metrics and can use tools such as ping and Packet Tracer to identify problems. I can analyze delay, packet loss, throughput, and congestion."

34. Explain a networking concept you are strong in.

Answer:

I am strong in QoS. QoS prioritizes important traffic such as voice and video. It helps reduce delay, jitter, and packet loss and improves the quality of real-time communication.

35. How would you handle a question you don't know?

Answer:

I would honestly say that I don't know the answer completely. I would explain what I know about the related concept and learn the correct answer afterward.

36. Describe a situation where you solved a technical problem.

Answer:

While working with a network simulation, if communication failed, I checked the IP addresses, subnet masks, connections, and routing configuration step by step. By identifying the incorrect configuration, I could restore communication.

37. Why should we select you for a networking role?

Answer:

I have a good foundation in networking concepts, I am willing to learn, and I can analyze problems systematically. I also have practical experience with network simulation tools.

38. What challenges did you face while learning Packet Tracer?

Answer:

Initially, configuring IP addresses, routing, and understanding packet flow was challenging. I overcame this by practicing different topologies and troubleshooting configurations step by step.

39. How do you improve your technical skills regularly?

Answer:

I improve my skills by practicing programs and simulations, studying concepts, solving problems, and learning from mistakes.

40. Explain a complex concept simply with confidence.

Answer:

For example, congestion is like a traffic jam on a road. When too many vehicles use the road, they move slowly and some may not reach their destination. Similarly, when too much data enters a network, delay and packet loss increase.